

Lab 03: Building a Wind Turbine

Physics and Mathematics of Sustainable Energy

Friday, October 2, 2026

College of the Atlantic

The **goals** for today are for you to:

- Gain a further understanding/appreciation for how electricity is generated.
- Have fun and feel a sense of triumph when the turbine turns.
- Optionally: Learn more about alternating current and electricity.

Guidelines

- Please work in pairs, in the same group as last week, if at all possible.

1. Find the generator you built last time.
2. Turn your generator into a wind turbine by making turbine blades. There are a variety of materials to choose from. You will be faced with some decisions: how many blades? what size and shape? what material? Discuss and strategize, and then build. (When using the utility knife, please be careful not to cut yourself. Knives are sharp. Also, please don't slice up the tables. Thanks.)
3. Optional, but recommended if you're in the mood: decorate your wind turbine.
4. Test your wind turbine in the "wind tunnel".
5. If time/interest, (and if I can get it working), we can experiment with the effects of different loads on the wind turbine, and also look at effects of turbine spacing.
6. If time/interest, I could show you how to use a multimeter to measure current and voltage.
7. Please check with me before you go.